Week 8 Notes

June 18, 2026

1 Goals

We already validated our VQE sweep against ED. Now we need to see if the circuit could handle realistic quantum noise effects.

To check this we start with using our noiseless VQE as the reference curve and then add noise in controlled stages. Then we can see effects of noise on different levels of the simulation and finally infer a boundary where our trends become unreliable.

2 Slide 4 - What noise can Break

Two major catogories. We can have errors in the state preperation on the measurement.

2.1 State Preperation

We use perfect rotation matrices in our code in physical hardware, this is a microwave pulse. If there is thermal fluctuation in the fridge, or if the pulse duration is slightly off by picoseconds, the qubit rotates by $\theta + \delta$. Your state vector $|\psi\rangle$ drifts away from the target ground state.

The Majorana signal depends on the first qubit (q_0) and the last qubit (q_{L-1}) being quantum-mechanically linked across the whole chain. This link is built by a "staircase" of CNOT gates. CNOTs are 10x to 100x noisier than single-qubit gates. If one CNOT fails to entangle properly, the "edge-to-edge" communication is severed, and the non-local string value will plummet.

Relaxation can bias the parity sector. This refers to T_1 relaxation ($|1\rangle \to |0\rangle$). In our Jordan-Wigner mapping, a $|1\rangle$ represents an occupied fermionic site. If a qubit "relaxes" to $|0\rangle$, it is physically equivalent to a particle escaping the chain. This changes the Fermion Parity from Even to Odd (or vice-versa). Since we rely on a strictly parity-constrained sector for our diagnostic, this "poisoning" ruins the reliability of the ground state.

2.2 Measurement

Our string order parameter is a product of eigenvalues: $X_0 \cdot Z_1 \cdot Z_2 \cdot X_3$. If the hardware misreads a single 0 as a 1 (a bit flip), the entire product changes sign (e.g., from $+1$ to -1). Because the string is a long product, the probability of the whole string being wrong increases with the system size L.

Shot noise broadens the string estimator: This is statistical variance. Because we only take a finite number of shots ($N = 4096$), our calculated average has an error bar $\sigma \approx 1/\sqrt{N}$. In a noisy environment, this "statistical jitter" makes the smooth curve of the phase diagram look like jagged noise.

A small true signal becomes hard to distinguish from zero: Near the phase boundaries ($|\mu| \approx 2t$), the Majorana edge modes are barely separated, and the "string" signal is naturally weak (maybe 0.1 or 0.2). If the noise floor of your hardware is 0.15, you can no longer tell if you are seeing a real topological signal or just hardware noise.

The "Main Risk": Contrast Flattening. This is the "big picture" failure. The goal of our project is to see a sharp distinction:Topological: String Value ≈ 1.0 and Trivial: String Value ≈ 0.0Noise acts like a "contrast slider" in a photo editor. It pulls the 1.0 down and the 0.0 up. If the noise is high enough, both phases will look like 0.4, and your diagnostic tool becomes useless. You "lose the phase boundary," which means your quantum computer can no longer distinguish between a topological superconductor and a regular piece of metal.

3 Two Experiments to keep seperate

f we introduce noise and run the whole VQE loop, and our final edge-string signal is garbage, we face a "chicken or egg" problem:Possibility A: The physical quantum gates and measurements ruined the state.Possibility B: The noise simply confused the classical optimizer so badly that it chose the wrong θ parameters.By doing a frozen-parameter noise test , we use the perfect $\theta^*(\mu)$ parameters from the noiseless Week 7 sweep and tell the classical optimizer to stay out of it.

4 Explicit forms of the noise matrices

Here are the explicit mathematical forms for the three completely positive trace-preserving (CPTP) maps defined in the presentation. Because the total system is 4 qubits, the full density matrix ρ is a 16×16 matrix, but these noise channels act on specific subspaces. 1. Gate Noise ($\mathcal{E}_{gate}$): The 2-Qubit Depolarizing ChannelThe presentation specifies that this noise is applied locally after each 2-qubit entangling gate. Therefore, this map acts on a 4×4 density matrix representing the 2-qubit subspace involved in the gate. A quantum channel is explicitly represented by a set of Kraus matrices $\{K_i\}$ such that $\mathcal{E}(\rho) = \sum K_i \rho K_i^\dagger$, subject to the trace-preserving condition $\sum K_i^\dagger K_i = I$.Using the presentation's parameterization (swapping their ρ for p to denote probability), the channel is constructed from 16 explicit Kraus matrices:The "No-Error" Matrix (1 matrix):

$$K_0 = \sqrt{1-p} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \sqrt{1-p}(I \otimes I)$$

The "Error" Matrices (15 matrices):For every non-identity 2-qubit Pauli operator $P_i \in \{IX, IY, IZ, XI, XX, \dots, ZZ$ the corresponding Kraus matrix is:

$$K_i = \sqrt{\frac{p}{15}} P_i$$

When these 16 matrices are applied to the 2-qubit state, they perfectly recreate the channel:

$$\mathcal{E}_{gate}(\rho) = (1-p)\rho + \frac{p}{15} \sum_{i=1}^{15} P_i \rho P_i$$

2. Basis Rotation ($\mathcal{E}_{meas-basis}$): The Unitary MappingThe slide defines this as an "ideal unitary mapping" to measure the edge string $X_0 Z_1 Z_2 X_3$. Quantum hardware typically only measures in the computational Z-basis. To measure X on qubits 0 and 3, the circuit must apply a basis-changing

unitary matrix U to rotate the X observable into the Z observable ($HXH = Z$). The required unitary matrix U applies a Hadamard gate (H) to qubits 0 and 3, and the Identity (I) to qubits 1 and 2.

$$U = H \otimes I \otimes I \otimes H$$

Since $H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, U is explicitly a 16×16 symmetric unitary matrix. The quantum channel is the standard unitary conjugation acting on the full state:

$$\mathcal{E}_{meas-basis}(\rho) = U\rho U^\dagger$$

3. Readout Assignment ($\mathcal{E}_{readout}$): The Stochastic MatrixThe presentation models readout error as an asymmetric classical bit-flip channel applied to the diagonal elements of the density matrix. Because measurement collapses the state, the diagonal of the 16×16 density matrix represents the 16 classical probabilities of measuring each bitstring (from 0000 to 1111). This channel is represented by a 16×16 classical transition matrix A_{full} acting on the probability vector $\vec{P} = \mathrm{diag}(\rho)$.First, we define the 2×2 assignment matrix for a single qubit. Let $p(0|1)$ be the probability of falsely reading a 0 when the true state is 1, and $p(1|0)$ be the probability of reading a 1 when the true state is 0. The single-qubit stochastic matrix A is:

$$A = \begin{pmatrix} 1 - p(1|0) & p(0|1) \\ p(1|0) & 1 - p(0|1) \end{pmatrix}$$

Assuming readout errors are independent across the 4 qubits, the explicit 16×16 matrix acting on the system is the tensor product of the individual qubit matrices:

$$A_{full} = A_0 \otimes A_1 \otimes A_2 \otimes A_3$$

The final map replaces the diagonal of ρ with the vector resulting from $A_{full} \cdot \mathrm{diag}(\rho)$. If the readout error is symmetric (where $p(1|0) = p(0|1) = \epsilon$), this is the exact mathematical mechanism that causes the edge string observable to shrink by a factor of $(1 - 2\epsilon)^4$.

5 On the form of the depolarising matrix

So essentially we do not change our matrix for 4 qubit system. I asked Gemini and the response was the following:

The short answer is: No, the core fraction $\frac{p}{15}$ does not change, but how we apply those matrices to the $L = 4$ system absolutely does.Here is the exact distinction between the local noise model your presentation uses, and a global noise model, which clarifies why the formula stays the same but the matrices grow.Global Noise vs. Local NoiseIf the presentation had assumed Global Depolarizing Noise (meaning a single error event randomly scrambles all 4 qubits at once), the formula would change.For an $L = 4$ system, there are $4^4 = 256$ total Pauli operators.Subtracting the Identity leaves 255 possible errors.A global noise channel would be: $\mathcal{E}_{global}(\rho) = (1-p)\rho + \frac{p}{255}\sum_{i=1}^{255} P_i \rho P_i$However, Slide 6 explicitly says: "Modeled via local depolarizing channels after each 2-qubit gate."This means the physical assumption is that the quantum hardware is mostly stable. The errors don't happen to the whole chip at once; they happen specifically to the two qubits that are currently interacting inside an entangling gate (like a CNOT). The other two "spectator" qubits are assumed to be resting perfectly safely.

Mathematically, the total gate noise map $\mathcal{E}_{gate}$ is actually a composition of three separate CPTP maps, one for each active pair:

$$\mathcal{E}_{gate}(\rho) = \mathcal{D}_{(2,3)} \circ \mathcal{D}_{(1,2)} \circ \mathcal{D}_{(0,1)}(\rho)$$

Here is exactly how that sequential timeline plays out mathematically during the simulation:The Sequential Noise TimelineStep 1: The (0, 1) GateThe hardware fires a microwave pulse to entangle Qubit 0 and Qubit 1. Right after this gate finishes, we apply the first depolarizing channel.We take the ideal state ρ and calculate a new, slightly corrupted state ρ_1.Note: Qubits 2 and 3 were resting safely during this.Step 2: The (1, 2) GateThe hardware fires the next pulse to entangle Qubit 1 and Qubit 2. We apply the second depolarizing channel.Crucially, this channel doesn't act on the perfect state; it acts on ρ_1 (which might already contain an error on Qubit 1). This creates an even more degraded state, ρ_2.Step 3: The (2, 3) GateThe final gate in the layer fires between Qubit 2 and Qubit 3. We apply the third depolarizing channel to ρ_2.This yields our final state for that circuit layer: ρ_3 (which is what the Python code ultimately outputs as `rho_gate`).

6 On the origin of δ_{SPAM}

On real quantum hardware, readout error is almost never symmetric. Because qubits lose energy to their environment (T_1 relaxation), a qubit in the $|1\rangle$ state is much more likely to erroneously decay to $|0\rangle$ during the measurement window than a $|0\rangle$ state is to spontaneously absorb heat and jump to $|1\rangle$.Here is the explicit mathematical proof of what happens when these probabilities diverge.1. The Asymmetric Transition MatrixLet's define the two distinct error rates:$\alpha = p(1|0)$: The probability of a false 1 (thermal excitation).$\beta = p(0|1)$: The probability of a false 0 (energy relaxation).Our single-qubit classical transition matrix A now looks like this:

$$A = \begin{pmatrix} 1-\alpha & \beta \\ \alpha & 1-\beta \end{pmatrix}$$

2. The Loss of the EigenvectorWe want to find the noisy expectation value $\langle Z\rangle_{noisy} = (A^T\vec{z})\cdot\vec{p}$, where $\vec{z} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\vec{p} = \begin{pmatrix} p_0 \\ p_1 \end{pmatrix}$.Let's see what happens when the transposed matrix A^T acts on the measurement vector $\vec{z}$:

$$A^T\vec{z} = \begin{pmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1-2\alpha \\ -1+2\beta \end{pmatrix}$$

Because $\alpha \neq \beta$, the result is no longer proportional to $\vec{z}$. The measurement vector is no longer a clean eigenvector of the noise.To solve this, we must mathematically split this new vector into two separate components—one proportional to the signal ($\vec{z}$), and one proportional to a constant offset ($\vec{i} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$):

$$\begin{pmatrix} 1-2\alpha \\ -1+2\beta \end{pmatrix} = (1-\alpha-\beta)\begin{pmatrix} 1 \\ -1 \end{pmatrix} + (\beta-\alpha)\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

3. The Two Effects of Asymmetric NoiseWhen we take the dot product of this split vector with the probability state $\vec{p}$, we get two distinct terms:

$$\langle Z\rangle_{noisy} = (1-\alpha-\beta)(\vec{z}\cdot\vec{p}) + (\beta-\alpha)(\vec{i}\cdot\vec{p})$$

We know that $\vec{z}\cdot\vec{p} = \langle Z\rangle_{ideal}$. Furthermore, $\vec{i}\cdot\vec{p}$ is simply $p_0 + p_1$, which must always equal exactly 1 (conservation of probability).This leaves us with the final equation for asymmetric single-qubit readout noise:

$$\langle Z\rangle_{noisy} = (1-\alpha-\beta)\langle Z\rangle_{ideal} + (\beta-\alpha)$$

Notice that asymmetric noise does two entirely different things simultaneously:Attenuation (The Scaling Factor): The signal is scaled down by $(1 - \alpha - \beta)$. This is the expected loss of contrast.Systematic Bias (The Shift): A constant term $+(\beta - \alpha)$ is added to the result, independent of the quantum state.4. Connection to the Kitaev Chain Failure CriterionBecause relaxation (β) is typically greater than excitation (α), the term $(\beta - \alpha)$ is positive. This means asymmetric readout noise physically creates a positive baseline drift in your parity measurements.When you scale this up to the 4-qubit string $Z_0 Z_1 Z_2 Z_3$, the math expands into a polynomial that not only scales down the ideal curve but physically shifts the entire bell curve up the y-axis.This perfectly explains the failure criterion equation from Slide 8:

$$\max_{\mu} |\langle \hat{O}_{edge}(\mu) \rangle_{meas}| \leq \frac{3}{\sqrt{N_{shots}}} + \delta_{SPAM}$$

The authors knew that simply scaling the curve wasn't the only risk. The δ_{SPAM} term explicitly accounts for the fact that asymmetric readout errors (State Preparation And Measurement errors) will artificially shift the measured parity up, making it look like there is a topological signal even in the trivial vacuum state where the true expectation value is zero.

7 On the difference between the Decoherence error and T_1 relaxation error

This difference is exactly why your presentation separates them. They break your VQE in distinct ways.The Depolarizing Effect: When depolarizing noise hits your circuit, it randomly flips bits and phases. Over the course of the simulation, this washes out the contrast of your edge string ($X_0 Z_1 Z_2 X_3$), pulling the expectation value toward 0. However, because the flips are symmetrical, the average number of particles in your Kitaev chain stays roughly the same.The T_1 Effect: Amplitude damping is uniquely lethal to parity-constrained simulations. Because T_1 physically changes $|1\rangle$ states into $|0\rangle$ states, it is literally deleting simulated "fermions" from your Kitaev chain. This breaks the parity conservation of your model. Instead of just washing out the signal to zero, T_1 relaxation introduces a massive systematic bias (that δ_{SPAM} drift), shifting your measurements into entirely wrong energy sectors that the ideal Hamiltonian wouldn't normally allow.

8 Why do we not consider T_1 and T_2 relaxation errors in our graphs

In Noisy Intermediate-Scale Quantum (NISQ) devices (like IBM's superconducting processors), different errors operate on vastly different timescales:Gate Times: A 2-qubit entangling gate takes roughly 10 to 100 nanoseconds. The operational error rate of this gate (depolarizing noise) is usually around 10^{-2} (1%).Coherence Times: T_1 relaxation and T_2 dephasing happen over 10 to 100 microseconds.Because the coherence times are thousands of times longer than the gate times, the probability of a qubit spontaneously decaying (T_1) or dephasing (T_2) during the execution of a shallow VQE circuit is mathematically dwarfed by the sheer probability of the entangling gates just messing up the operation (depolarizing). By focusing on depolarizing and readout errors, the authors are isolating the dominant bottlenecks that will kill the signal first.